

AD 502 – Machine Learning

Pre-Reading Material – Lecture 20

Partitioning Clustering

Why Do We Study Clustering?

Clustering is an **unsupervised learning technique** used to group similar data objects together.

Unlike classification, clustering works with **unlabeled data**. The algorithm tries to discover natural groups or patterns present in the dataset.

Applications

- Customer segmentation
- Document and news grouping
- Image segmentation
- Market analysis
- Recommendation systems
- Biological data analysis

What is Partitioning Clustering?

Partitioning clustering divides a dataset into a predefined number of groups called **clusters**.

Each data object is generally assigned to **one cluster**, so the dataset is partitioned into separate groups.

The main objective is to create clusters such that:

- Objects within the same cluster are **similar**.
- Objects belonging to different clusters are **different**.



Basic Idea

Suppose a dataset contains many customers with different characteristics.

A partitioning algorithm can divide them into groups such as:

Customer Data → Clustering Algorithm → Cluster 1, Cluster 2, Cluster 3, ...

The number of clusters is usually specified before the algorithm starts.

This number is commonly represented by **K**.

K-Means Clustering

K-Means is one of the most commonly used partitioning clustering algorithms.

Its basic idea is to represent each cluster using a **centroid**, which represents the center of that cluster.

Basic Steps

1. Choose the number of clusters (**K**).
2. Initialize the cluster centroids.
3. Assign each data point to the nearest centroid.
4. Recalculate the centroids using the assigned points.
5. Repeat the assignment and centroid-update steps.
6. Stop when the clusters become stable or the stopping condition is reached.

Example of K-Means

Consider customer data based on:

- Annual income
- Spending behavior

K-Means may identify groups such as:

- **Cluster 1:** Low income, low spending
- **Cluster 2:** High income, high spending

- **Cluster 3:** High income, low spending

The algorithm discovers these groups from the data rather than using predefined class labels.

Similarity and Distance

Partitioning algorithms need a way to determine how similar or different two data points are.

Common distance measures include:

- **Euclidean distance**
- **Manhattan distance**

The selected distance measure can influence the resulting clusters.

Choosing the Number of Clusters

A major question in partitioning clustering is:

How many clusters should be created?

One commonly used approach is the **Elbow Method**.

It examines how the clustering error changes as the number of clusters increases.

The aim is to identify a point where adding more clusters provides relatively little additional improvement.

Advantages of Partitioning Clustering

- Simple and easy to understand.
- Computationally efficient for many datasets.
- Works well when clusters are reasonably well separated.
- Easy to implement and interpret.

Limitations

- The number of clusters usually needs to be specified.
- Results can depend on the initial centroid selection.
- Sensitive to outliers.
- K-Means may not perform well when clusters have irregular shapes or significantly different sizes.

Learning Outcomes

After studying this topic, you should be able to:

1. Explain the concept of partitioning clustering.
2. Differentiate clustering from classification.
3. Describe the working of K-Means clustering.
4. Explain the role of centroids and distance measures.
5. Understand how the number of clusters can be selected.

Think Before the Lecture

1. How is clustering different from classification?
2. Why does K-Means require the value of **K**?
3. What happens if a data point is equally close to two cluster centroids?
4. Why can the initial selection of centroids affect the final clusters?
5. Why might K-Means struggle with irregularly shaped clusters?

Quick Recall

Data → Choose K → Initialize Centroids → Assign Points → Update Centroids → Repeat → Final Clusters

Key idea: Partitioning clustering divides data into groups so that objects within a cluster are more similar to each other than to objects in other clusters.